

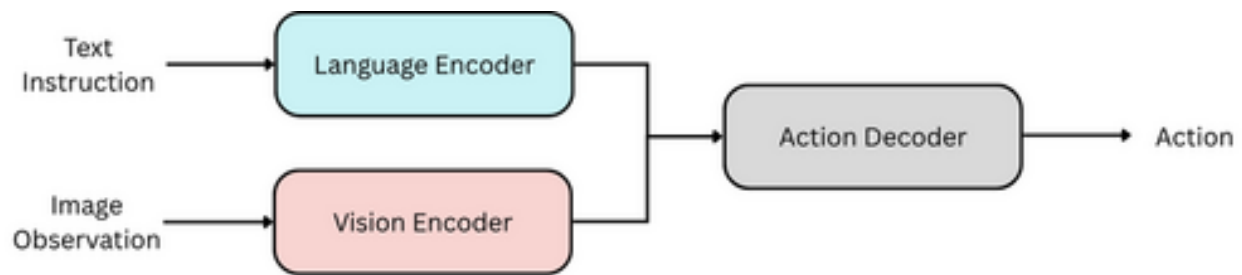
Project 1: Intelligent Pick-and-Place Robot Using a Vision-Language-Action Models

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Number of Students: 8

Objective

Recent advances in Vision–Language–Action (VLA) models have enabled robots to perform complex manipulation tasks by jointly reasoning over visual observations, natural language instructions, and action generation. As shown below, a typical VLA architecture consists of a vision encoder that extracts semantic information from the robot's visual observations, a language encoder that processes task instructions, and an action decoder that integrates these multimodal representations to generate a sequence of low-level control commands for robotic execution.



Given an input image (or video) of the robot's environment together with a natural language instruction, the VLA model directly predicts executable robot actions, allowing the robot to autonomously perform the specified task. This unified perception–reasoning–action paradigm represents a significant step toward the development of general-purpose intelligent robotic systems capable of operating in unstructured environments.

The objective of this project is to design, develop, and implement an autonomous robotic manipulation system based on the Vision–Language–Action paradigm. The proposed system will integrate a robotic arm with vision sensors and a VLA model to enable object detection, scene understanding, task interpretation, motion planning, and object manipulation from natural language instructions. The implementation may initially be validated in a high-fidelity simulation environment, such as NVIDIA Isaac Sim, before deployment on a physical robotic platform. The project will investigate the effectiveness of VLA models in performing autonomous pick-and-place and object manipulation tasks while evaluating system performance in terms of task success rate, inference efficiency, robustness, and

generalization to previously unseen objects and environments. This is suitable for **Mechatronics and Computer Engineering students covering the following topics:**

- Robot arm design
- Servo motor control
- End-effector design
- Motion planning
- Inverse kinematics
- Safety mechanisms
- Vision-Language-Action model
- Object detection
- Scene understanding
- Action planning
- Robot control software
- ROS2 integration

Hardware

- 6-DOF robotic arm (or xArm Lite / MyCobot / Niryo One)
- NVIDIA Jetson Orin Nano
- Intel RealSense D435 or Luxonis OAK-D camera
- Servo motors
- Gripper
- ROS2-compatible controller

Software

- ROS2
- MoveIt2
- NVIDIA Isaac Sim

- PyTorch
- OpenVLA or Pi0/OpenVLA-compatible models
- Segment Anything (SAM)
- YOLOv11
- TensorRT

AI Components

- Object detection
- Scene segmentation
- Language understanding
- Grasp planning
- Motion generation
- Closed-loop visual feedback

Expected Deliverables

- Voice-controlled robot arm
 - Natural language interface
 - Automatic object manipulation
 - Digital twin in Isaac Sim
 - Real-world validation
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